

NEW HAVEN HARBOR, CT, USA

WMO: 997284

Lat: 41.283N

Lon: 72.908W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.3	(c) -9.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 12.8	(k) -3.2	(l) 12.1	(m) -2.8	(n) 5.5	(o) 310	(p) 0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.8	(c) 28.9	(d) N/A	(e) 27.5	(f) N/A	(g) 26.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.5	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.5	9.0	-13.7	32.4	2.1	1.6	-15.2	33.5	-16.4	34.5	-17.6	35.4	-19.1	36.5
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-0.9	0.0	3.8	9.8	14.9	20.0	23.3	22.5	19.1	13.5	7.2	3.1
	DBStd	9.11	4.82	4.25	4.01	3.49	3.18	2.97	2.38	2.19	3.12	3.65	4.01	4.30
	HDD10.0	1187	337	279	194	45	2	0	0	0	0	11	101	218
	HDD18.3	2936	596	513	450	257	113	17	1	1	29	155	333	473
	CDD10.0	1707	0	0	3	38	155	299	412	387	272	119	18	3
	CDD18.3	414	0	0	0	0	8	66	154	129	51	5	0	0
	CDH23.3	1675	0	0	0	1	19	189	756	561	138	11	0	0
	CDH26.7	222	0	0	0	0	4	29	124	59	4	2	0	0
Wind	WSAvg	4.1	4.4	4.5	4.7	4.1	3.8	3.8	3.7	3.4	3.7	4.1	4.4	4.3
Precipitation	PrecAvg	1065	74	79	89	98	99	88	88	80	86	90	101	96
	PrecMax	1457	155	135	234	272	268	322	222	199	196	211	220	196
	PrecMin	703	6	14	18	36	20	6	21	18	23	7	9	26
	PrecStd	212	40	28	46	52	62	72	54	46	52	52	59	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	11.6	15.8	22.4	25.5	29.8	31.9	30.2	27.4	24.6	17.9	14.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	9.7	8.5	12.8	18.8	22.8	27.1	29.4	28.1	26.1	21.4	16.0	12.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	7.7	7.0	10.9	16.6	21.1	25.4	27.8	27.0	25.0	20.2	15.0	11.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	5.9	5.6	9.3	14.8	19.7	24.0	26.6	26.1	23.9	19.0	13.7	9.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.6	6.5	7.0	6.8	6.3	5.8	6.1	6.8	6.8	6.6	5.8
		MCDBR	8.2	7.9	8.4	9.7	8.5	7.7	7.1	6.7	6.4	7.3	7.1	6.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.308	0.317	0.341	0.376	0.425	0.470	0.487	0.461	0.389	0.349	0.319	0.306	
	taud	2.472	2.452	2.439	2.372	2.264	2.177	2.139	2.205	2.379	2.507	2.543	2.538	
	Ebn at Noon	848	895	910	895	854	813	794	804	846	848	832	824	
	Edh at Noon	76	90	102	117	134	146	151	137	108	85	71	66	
All-Sky Solar Radiation	RadAvg	1.85	2.72	3.72	4.80	5.37	5.85	5.97	5.31	4.36	2.94	2.05	1.54	
	RadStd	0.15	0.25	0.29	0.51	0.53	0.57	0.32	0.32	0.36	0.28	0.17	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	+58

Nomenclature: See separate page